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GROUP 1: Rotational Motion to Linear Motion Machine

Mech 206 Project Report



Bertan Erman 83155

Can Maral 87053

Furkan Özgümüş 87023

Mustafa Başar Ezik 83468

Yiğit Berk Uçar 88211

TA : HASSAN AKBAR

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1. Introduction

1.1 What is Our Project?

This project presents the design and production of a mechanical system that converts a rotational input into an oscillating linear output. The mechanism aims to demonstrate the principles of motion transformation through linkages and gear systems.

1.2 What is It Used For?

The conversion of rotational motion into oscillating linear motion is widely used in engineering systems such as electric shavers, hammer drills, and electric saws. These devices often rely on compact and efficient linkages to transfer motor torque into useful mechanical systems. Our project serves as a simplified, yet functional prototype of such mechanisms.

1.3 How Does It Operate?

The rotational input is provided by a stepper motor, which drives a gear system. This gear transfers motion to an L-shaped bar, which in turn rotates the far point of the center rod in a circular trajectory. Due to the system's design, the vertical components of the center rod's motion are filtered out, and only the horizontal motion is transmitted to a centerpiece. This centerpiece is connected to two side rods via sliders, which restrict motion transmission to only the relevant axis, thereby generating oscillating linear output at the ends. The last gear in the train is directly connected to an angled arm that initiates the spinning of the center rod about the same axis as the driving gear. This designed motion pathway ensures single-axis oscillation.

(a) (b)

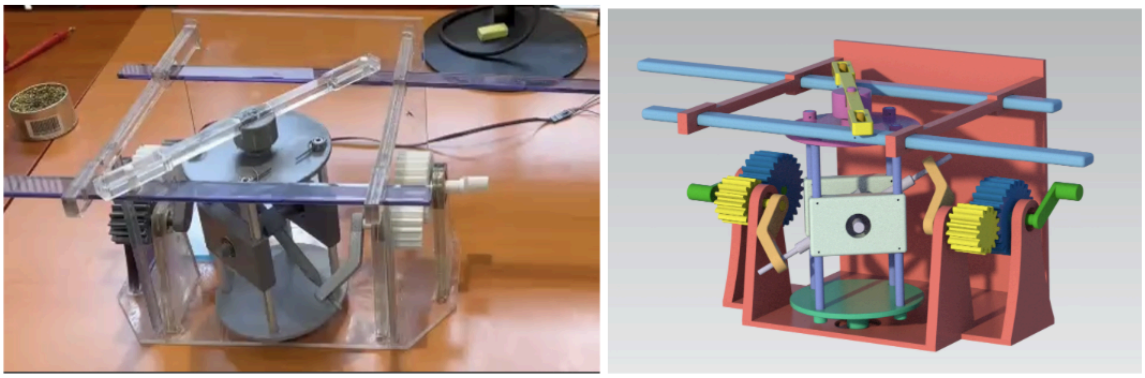


Fig 1. (a) Real photo of our design. (b) Render photo of our CAD design.

2. Technical Background

2.1 Mechanism in Detail

Our mechanism incorporates several fundamental mechanical elements:

1. A stepper motor that initiates motion.
2. A gear train, which transfers and adjusts torque and speed.
3. A rotating L-bar that imparts orbital motion.
4. A center rod whose horizontal components are transferred to a centerpiece.
5. Sliders and side rods that produce the final oscillating linear motion.

2.2 Linkages and Gear Systems

The project integrates bar linkages and rotational gear systems, specifically:

1. Four-Bar Linkage: The center of our system can be interpreted as a constrained four-bar mechanism, transmitting motion between fixed and moving rods. A four-bar linkage typically includes one grounded link, two rotating links, and one coupler, offering 1 DoF (Degree of Freedom) [1].

2. Gear Train: The gear system is composed of external spur gears that alter rotational direction and magnitude. These gears maintain constant velocity ratios while allowing torque distribution to the rotating arm.
3. Slider Mechanism: The sliders connecting the centerpiece and side rods constrain motion transmission only along a desired axis. This eliminates vertical noise in motion and improves control.

These mechanisms together create a hybrid motion system rotational at input, oscillating and linear at output.

2.3 Degrees of Freedom (DoF)

Our mechanism has a single degree of freedom (1 DoF). This means that once the stepper motor rotates, all other components move in a fully determined way. The Gruebler-Kutzbach criterion confirms this configuration.

2.4 Industrial / Potential Applications

This mechanism can be found or adapted for the following applications:

1. Electric shavers: oscillating blades powered by rotary motors.
2. Hammer drills: rotational energy converted into percussive motion.
3. Reciprocating saws: converting rotation into back-and-forth motion.
4. Automotive valves: cam-based linkage systems for opening/closing.

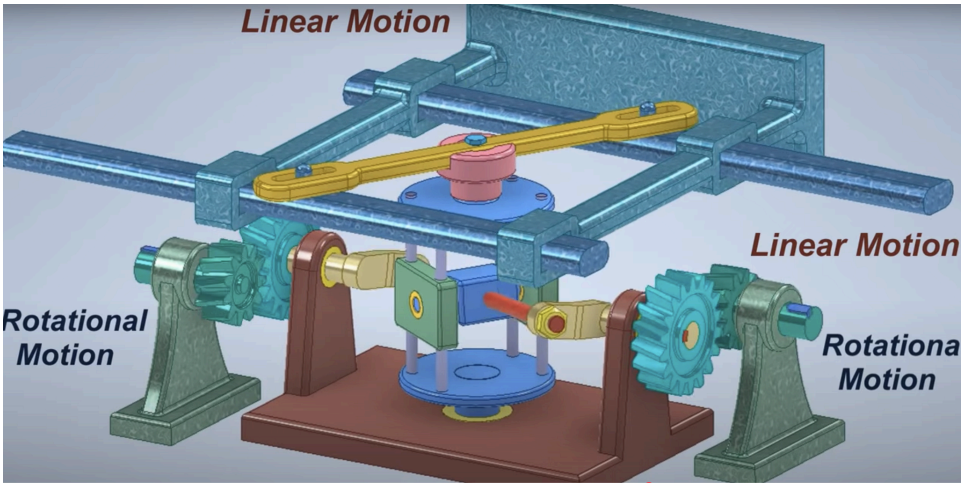


Fig 2. Another example of our system [2].

3. CAD Design

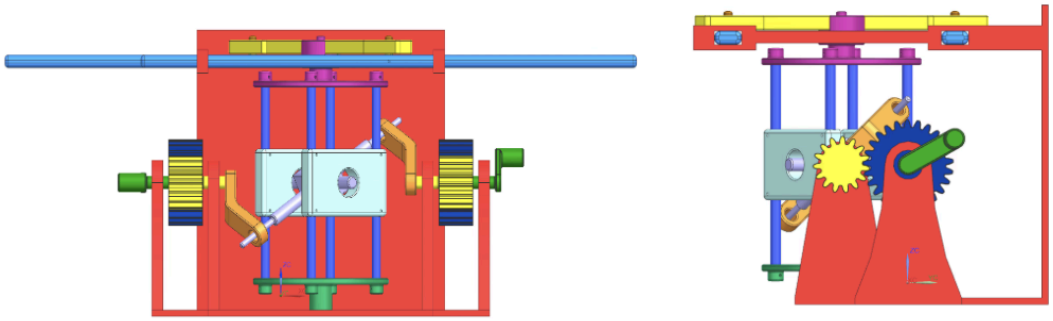
3.1 Description of the CAD Model

The CAD design was developed using SiemensNX. The model includes all parts, assemblies and drafts: the base, gears, rods, sliders, and housing parts. Some components were designed to be laser-cut or 3D printed, reflecting practical manufacturing constraints.

Special attention was given to mechanical fit, motion range, and tolerance compensation in the CAD model. Plexiglass parts were modeled with double-layered geometry (2 × 5 mm) due to limitations in the laser cutter.

3.2 Variations in the Market and Comparison

There are commercial versions of similar mechanisms in power tools. While these are often more compact or involve servo or DC motors and control systems, the fundamental mechanical behavior, converting rotational to linear motion, remains the same.



(a) (b)

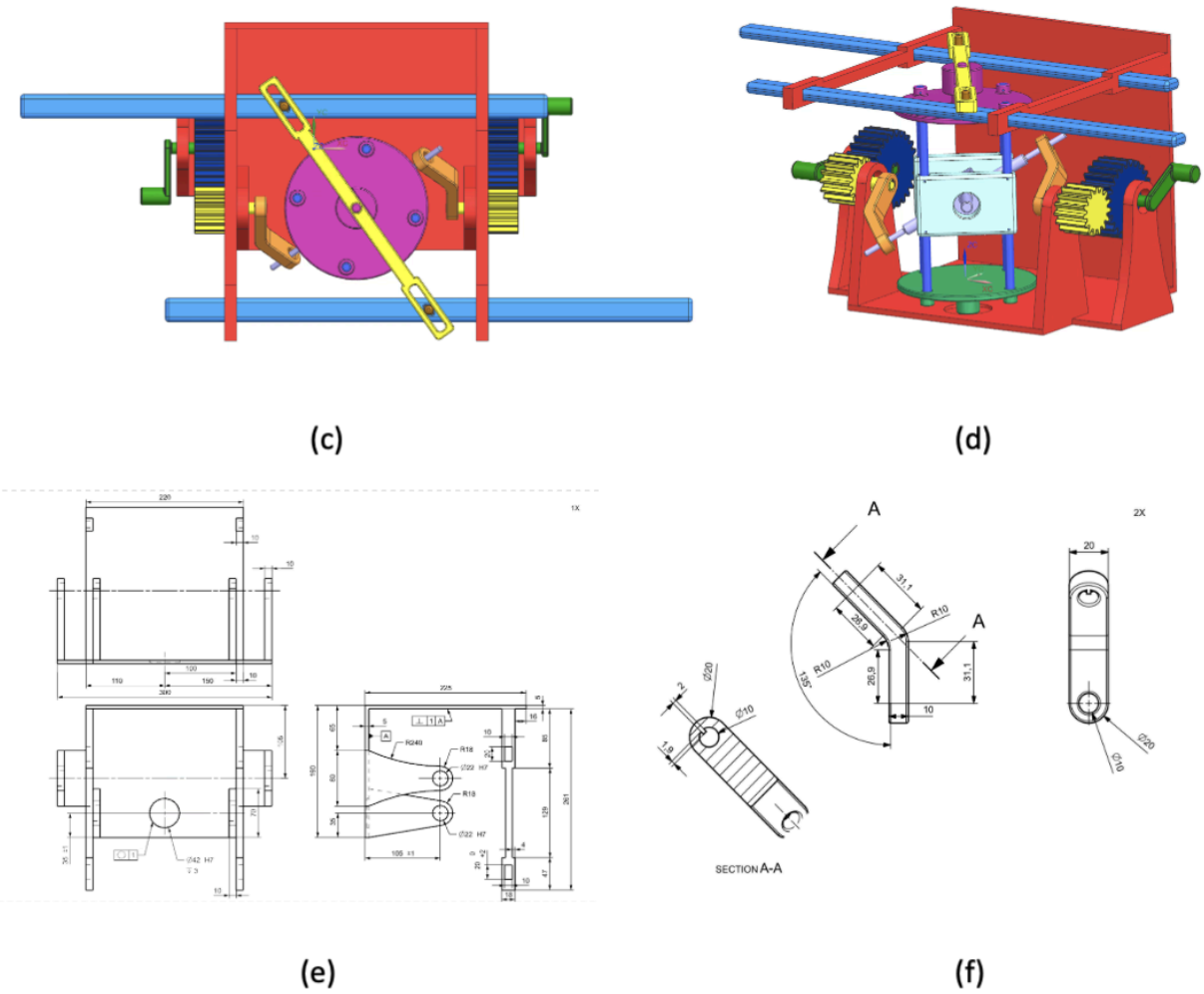


Fig 3. (a),(b),(c),(d) photos of our CAD design from different angles. (e),(f) Drafts of two important parts.

4. Mechanism Actuation and Manufacturing

In our gear design we used spur gears with 20 degree pressure angles which are very common and can be used in almost any rotational motion translation.

The base of the mechanism is made from plexiglass that was cut using a 2D laser cutting machine. To achieve the 3D part the parts that stuck out were made to fit and then glued together to ensure it won't come apart. The linear motion rods at the top were also made from plexiglass since they are 440 mm long so 3D printing would be more expensive and take longer.

The moving parts such as gears, the center parts, slider pins, and rods were made from 3D printing. Since they needed to be light and had complex shapes, 3D printing was the best option.

After the parts were made they were fit together with glue where necessary and bearings were used in 9 places to minimize friction losses.

When the manufacturing was finished we saw that the machine still had a lot of internal friction that made it run with inconsistent speed. We used liquid grease in the sliders and gears to resolve this problem and greasing the motion of the system was considerably smoother.

We have also integrated a motor driving mechanism using Arduino. Motor we have used is a stepper motor (17HS4402) which works at a 30 to 120 rpm (3.14-12.56 radian/s). Motor driver is (L298). Steps per revolution in code is set to 300. In one full revolution each step moves (360/300) = 1.2 degree. Additionally, motor speed can be controlled by rotary potentiometer.

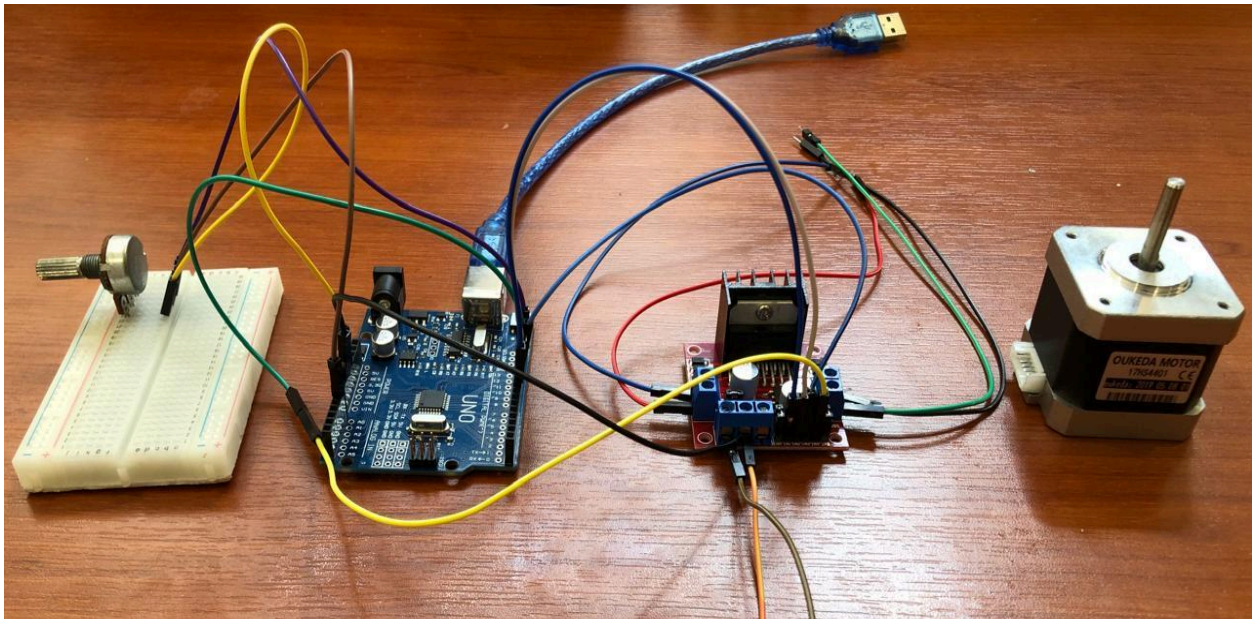


Fig. Establishment of the motor control circuit using Arduino

5. Bills of Materials

ITEM NUMBER	ITEM NAME	CATEGORY	PHASE	QTY	UNIT	SUPPLIER	COST (₺)	TOTAL (₺)	SOURCE LINK	BOM NOTES
ELECTRONICS										
ELEC-001	Arduino Uno R3	Control	Prototype	1	pc	Robotistan	186.08	186.08	https://www.robotistan.com/arduino-uno-r3-klon	Main controller
ELEC-002	NEMA17 Stepper Motor	Actuation	Prototype	1	pc	Robotistan	327.99	327.99	https://www.robotistan.com/nema-17-step-motor	45 Ncm torque
ELEC-003	L298N Motor Driver	Power	Prototype	1	pc	Robotistan	30	30	https://www.robotistan.com/l298n-motor-surucu-karti	Motor control
ELEC-004	10KΩ Potentiometer	Control	Prototype	1	pc	Robotistan	8.46	8.46	https://www.robotistan.com/10k-ohm-potansiyometre	Speed adjustment
MECHANICAL										
MECH-001	Spur Gear (Module 1)	Transmission	Production	2	pc	Robotistan	25	50	https://www.robotistan.com/disli-modul-1	20 teeth
MECH-002	Linear Bearing 8x22x7mm	Motion	Production	8	pc	Robotistan	8	64	https://www.robotistan.com/lineer-bilya-yatak	Flanged type
MECH-003	Aluminum Rod Ø8mm	Structure	Prototype	3	m	Local	20	60		Structural support
MATERIALS										
MAT-001	ABS Filament 1.75mm	3D Printing	Production	1	kg	Robotistan	250	250	https://www.robotistan.com/pla-1-75-mm	Gear manufacturing
MAT-002	Plexiglass 3mm	Structure	Production	1	m²	Local	150	150		Frame material
CONSUMABLES										
CONS-001	Jumper Wires	Electrical	Prototype	1	kit	Robotistan	44.64	44.64	https://www.robotistan.com/jumper-kablo	40pc 20cm

CONS-002	Breadboard 400pt	Electrical	Prototype	1	pc	Robotistan	33.38	33.38	https://www.robotistan.com/breadboard-400-delikli	Circuit prototyping
							→	1204.55		

6. Motion simulation

The base of the mechanism was assigned a fixed joint to establish a non-moving reference frame. Revolute joints were employed for both gears to enable rotational motion about their respective axes. A gear coupler was introduced between the gears, with the appropriate gear radii defined to ensure realistic gear interaction and torque transfer. An angled arm was rigidly attached to the smaller gear. To replicate the behavior of bearings located at the ends of the middle rod, revolute joints were applied. This choice was crucial; had fixed joints been used in locations with bearings, the model would have become over-constrained, preventing rotational motion. The middle body was also connected via revolute joints, allowing for free rotation. For the linear actuating components—namely the sliders—slider joints (also known as prismatic joints) were utilized, which constrained the motion to a single translational degree of freedom along the defined axis. To drive the mechanism, a constant angular velocity of 3.14 rad/s was applied to the smaller gear. The simulation was run under this input condition to evaluate the resulting angular and linear velocities and accelerations of the interconnected parts. This comprehensive joint setup ensures that both rotational and translational motions are accurately modeled, providing realistic system behavior under defined boundary conditions. Their kinematic dataset is exported to Excel data sheets to be used in MATLAB, where we compared experimental data with Siemens NX motion outputs. This allows for precise comparison between simulation results and physical system performance, facilitating validation and potential design improvements. Without proper constraint selection and joint assignment, mechanical simulations can easily become unrealistic or even unsolvable. Therefore, this setup was carefully designed to reflect the actual physical system with high fidelity.

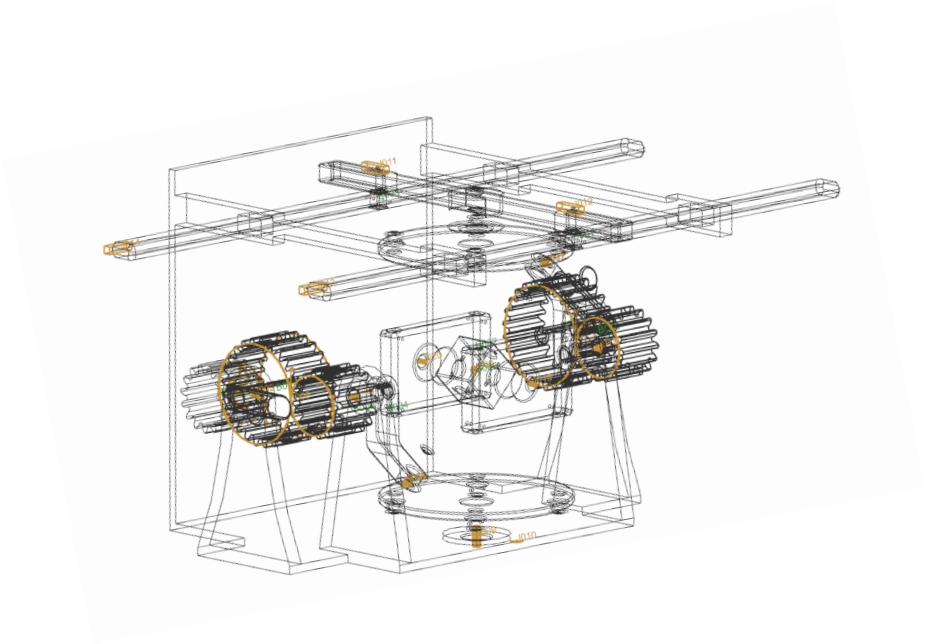


Fig 4. Motion Bodies, Links, Gear Couplers at Siemens NX with 100 percent translucency on parts

7. Derivation of Dynamic Equations

Description: Our machine translates rotational motion to linear motion and there are two possible supplies of our machine.

1) By a motor that is connected to the big gear by another gear

2) By a lever arm that is coaxial with the big gear.

Features of the Motor

Our motor is at the center of a gear which has the following values

Pressure angle = 20 degrees

Module(d) = 2.5 mm

Number of tooth (N_{LG})= 15

R_{SL} = 21.25 mm = Outer circle radius

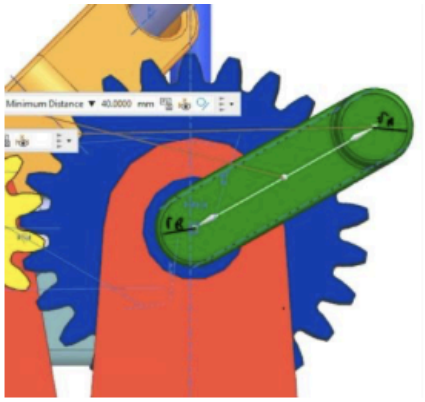
R_{PL} = 18.75 mm = Pitch circle radius

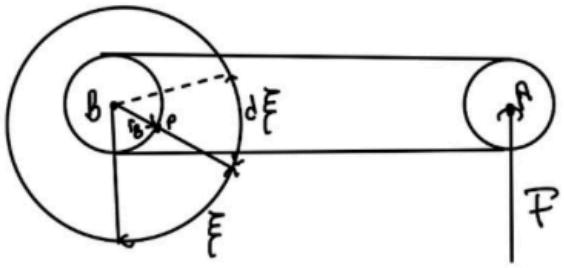
Features of the Green Lever

Distance between the centers of the circles (GL): 40 mm = 4 cm

Radius of the Circle A (ra): 7.5 mm = 0.75 cm

Radius of the Circle B (rB): 6 mm = 0.6 cm





Gears

There are 3 important inputs to determine the features of spur gears. These are the number of teeth, the module and the pressure angle. For gears that make contact, for the contact to be made smooth and frictionless as possible the gears pressure angle and module need to be the same.

Then the gears will touch at their pitch circle which is found by multiplying No. of teeth and module.

Features of the Big Gear

This gear is connected to the lever arm and their centers are co-axial according to turning axis.

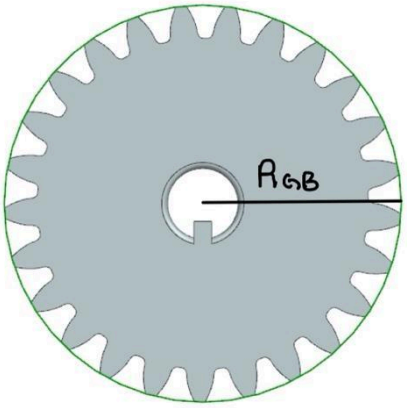
Pressure angle = 20 degrees

Module(d) = 2.5 mm

Number of tooth (N_{BG}) = 25

R_{BG} = 33.75 = Outer circle radius

R_{PB} = 31.25 = Pitch circle radius



Dynamics

Due to having co-axial center with the lever arm, so there will be the same amount of change in the angular position of this gear which is $d(\xi)$. Velocity at the pitch circle of the gear can be calculated with the following equation:

$$v_{BG} = \omega_{BG} \times R_{PB}$$

Acceleration can be found by the equation:

$$a_{BG} = a_{\omega} \times R_{PG} + \omega \times (\omega \times R_{PG})$$

This big gear has a connection with the little gear. At the point of that big gear touches the small gear their velocities will be the same.

Features of the Small Gear

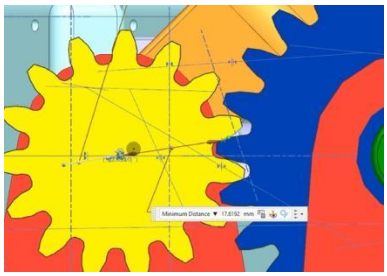
Pressure angle = 20 degrees

Module(d) = 2.5 mm

Number of tooth (N_{LG})= 15

R_{SL} = 21.25 mm = Outer circle radius

R_{PL} = 18.75 mm = Pitch circle radius



Dynamics

From the photos given below we can understand that tangential components of the gears have the same magnitudes. These tangential components can be calculated with the following equation:

$$(\omega_{BG})N_{BG} = -(\omega_{LG})N_{LG}$$

$$v_{LG} = \omega_{LG} \times R_{PL}$$

$$a_{LG} = \alpha_{\omega} \times R_{LG} + \omega_{LG} \times (\omega_{LG} \times R_{BG})$$

$$\alpha_{\omega BG} = -\alpha_{\omega LG}$$

Features of the Angled Arm

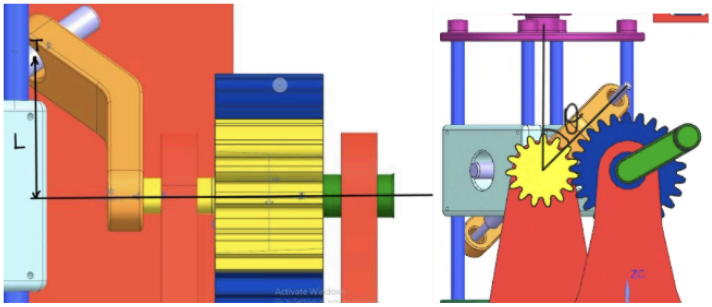
Inner angle of the arm: 45 degrees



Dynamics of the Angled Arm

Little gear is connected to the angled arm from the center so they are co-axial; thus, the angular velocities of the little gear and the angled arm. From the supplied photo it can be seen that the angled arm and the central stick are connected to each other. It can be said that the velocity at the center of the hole(in the arm) and shaft in the stick are the same at the fit center(Let's call it T).

We should find the distance between the point T and the projection of point T on the centerline of the little gear. Let's call this distance (L) and it is measured as 56.6 mm. So the displacement from the center to the angled arms holes center is R_T which has a magnitude of L.



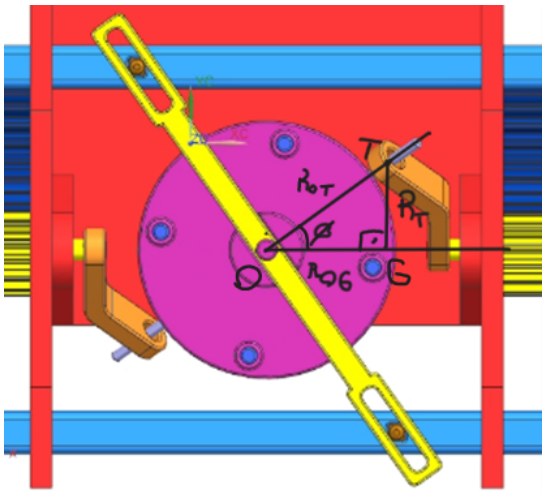
$$v_T = \omega_{LG} \times R_T$$

$$a_T = \alpha_{\omega T} \times R_T + \omega_{LG} \times (\omega_{LG} \times R_T)$$

$$\left(R_T\right)_y = \left|R_T\right| sin\,\theta$$

$$\left(v_T\right)_y = \left|v_T\right| cos\,\theta$$

$$\left(a_T\right)_y = \left|\alpha_{\omega T} \times R_T\right| cos\,\theta + \left|\omega_{LG}\right|\left|v_T\right| sin\,\theta$$



The change in y position of the T point when looked from this angle given above causes the motion of the centerpiece. The angle of rotation can be calculated using. (R is the length from the

$$\phi = arctan\left(\frac{\left(R_T\right)_y}{\left|R_{OG}\right|}\right)$$

Then the derivatives are:

$$\omega_\phi = \frac{\frac{\left(v_T\right)_y}{\left|R_{OG}\right|}}{1+\left(\frac{\left(R_T\right)_y}{\left|R_{OG}\right|}\right)^2} = \frac{\left(v_T\right)_y * \left|R_{OG}\right|}{\left(R_T\right)_y{}^2+\left|R_{OG}\right|^2}$$

$$\alpha_\phi = \frac{\left|R_{OG}\right|\left[\left(a_T\right)_y\left(\left(R_T\right)_y{}^2+\left|R_{OG}\right|^2\right)-2\left(R_T\right)_y\left(v_T\right)_y\right]}{\left(\left(R_T\right)_y{}^2+\left|R_{OG}\right|^2\right)^2}$$

At this picture R_J will be given by the formula

$$R_J = tan\phi * \left|R_{OJ}\right| = \frac{\left(R_T\right)_y}{\left|R_{OT}\right|} * \left|R_{OJ}\right|$$

$$v_J = \left(v_T\right)_y \frac{\left|R_{OJ}\right|}{\left|R_{OT}\right|}$$

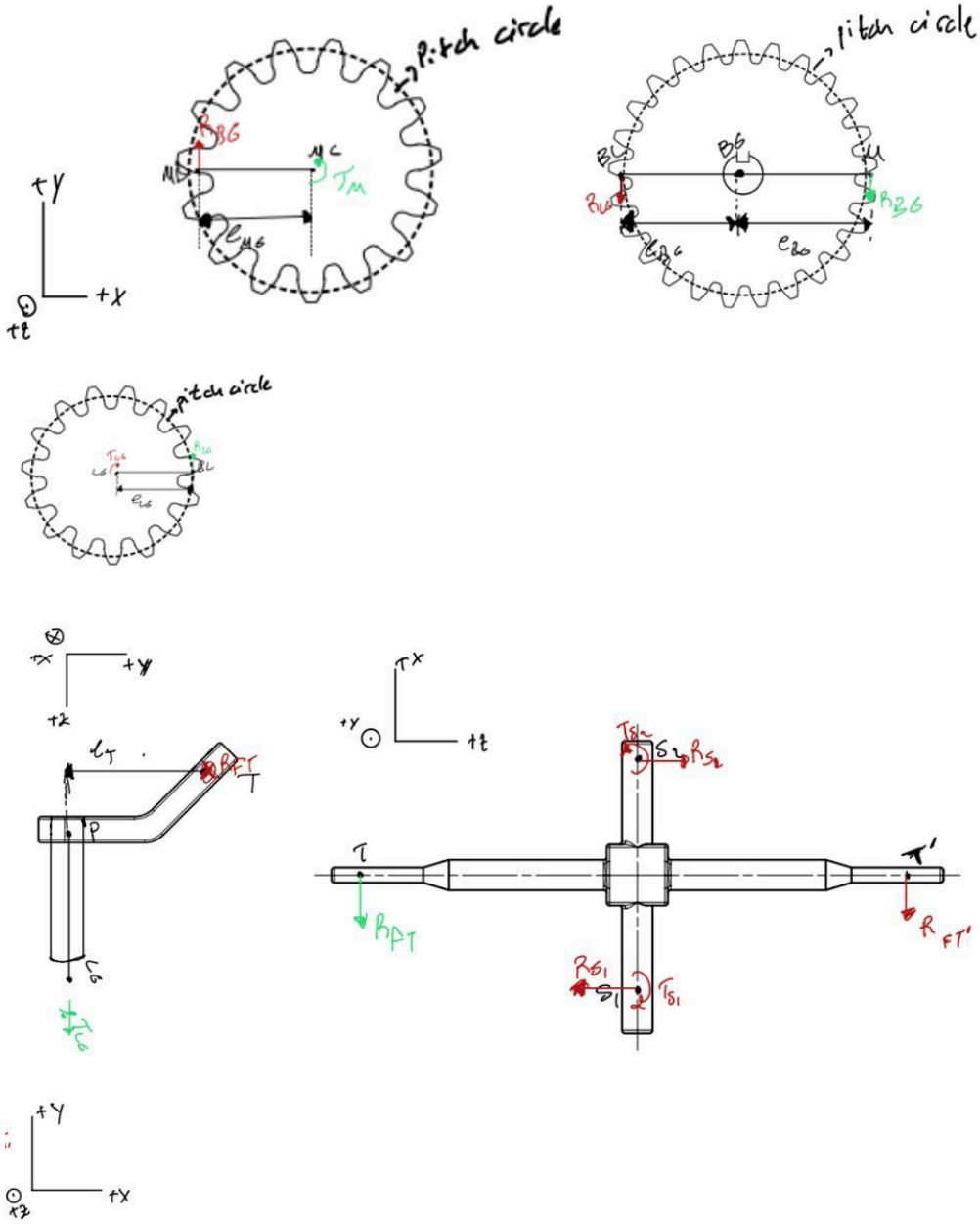
$$a_J = \left(a_T\right)_y \frac{\left|R_{OJ}\right|}{\left|R_{OT}\right|}$$

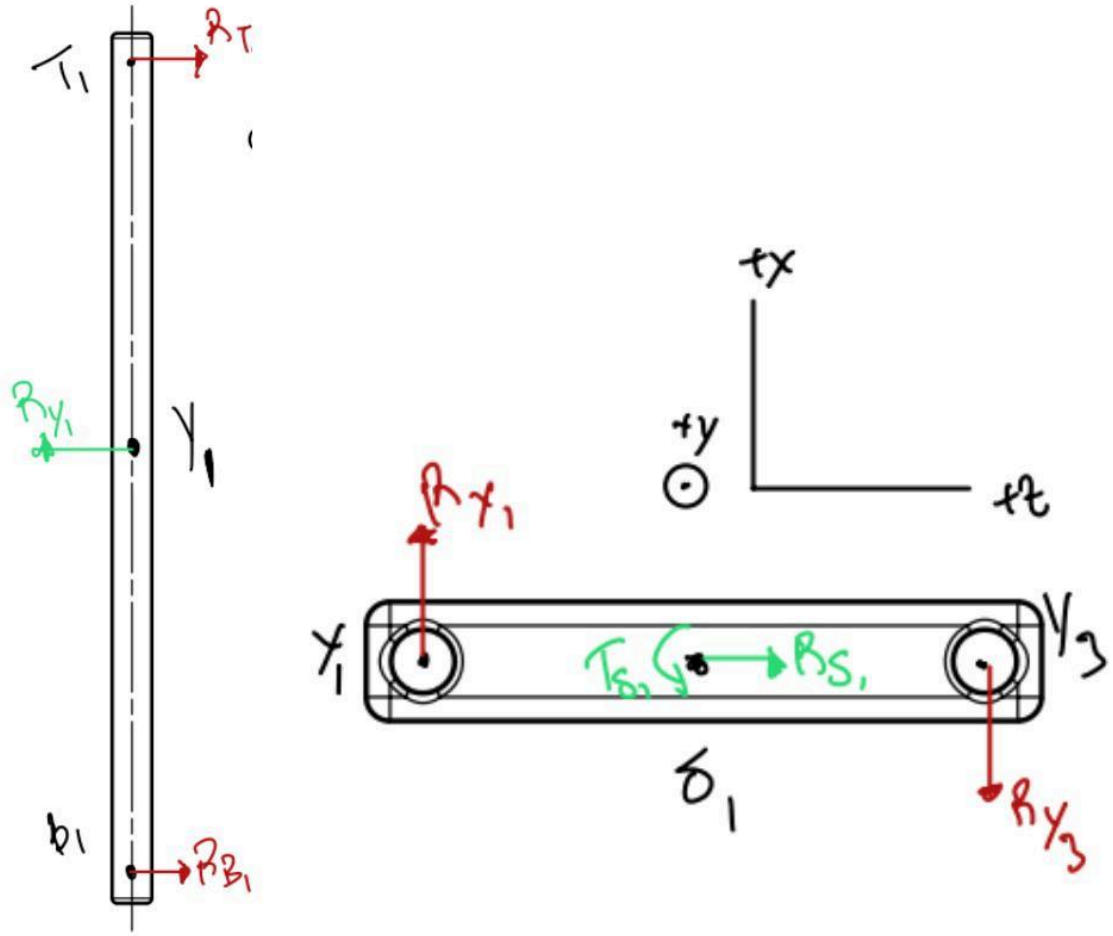
For given initial conditions of ω_{BG}, α_{BG} *and any angle to determine position such as* θ *or* ϕ

.

The system can be solved.

7.1. Force and Acceleration





7.2. Torque Calculation

It should be noted that electric motors do not operate under ideal conditions in practice, and energy losses occur. Therefore, the actual mechanical output power of the 17HS4402 motor is calculated by applying the system efficiency coefficient (η) to the electrical input power. In this study, it is assumed that the motor operates with 70% efficiency.

7.2.1 Electrical Power Calculation

Electrical power is calculated using Ohm's Law as follows:

$P_{electric} = I^2 \cdot R$

Where:

I = 1.2 A (current)

R = 2.5 Ω (resistance from datasheet)

$P_{electric} = (1.2)^2 \cdot 2.5 = 1.44 \cdot 2.5 = 3.6$ W

7.2.2 Actual Mechanical Power

Assuming the motor efficiency (η) is 70%, the output power is calculated as:

$P_{mechanical} = \eta \cdot P_{electric} = 0.70 \cdot 3.6 = 2.52$ W

7.2.3 Torque Calculation

Given that the angular velocity (ω) is 3.14 rad/s, the relationship between mechanical power and torque (T) is expressed as:

$T = P_{mechanical} / \omega$

T = 2.52 / 3.14 = **0.802 Nm**

7.3 Moment of Inertia Calculation

For the Moment of Inertia of the gears we took two different approaches

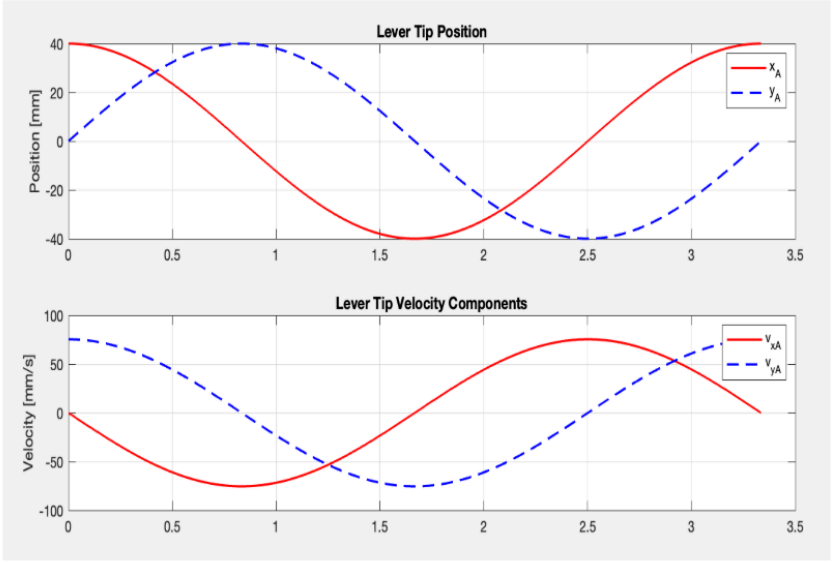
- Considering them as a cylinders $I_{ZZ} = (MR^2) * 1/2$
- Calculating the real value using Siemens Nx. The moment of inertia of the small gear calculated by assigning material (Polylactic Acid PLa is assigned and density is 0.00123 g/mm^3)

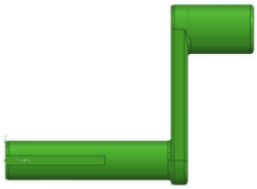
```
Inertias
        IXX=       54.95570 kg·mm²  IYY=       35.65274 kg·mm²  IZZ=       35.64945 kg·mm²
        IXY=        0.00000 kg·mm²  IXZ=        0.00000 kg·mm²  IYZ=        0.00000 kg·mm²
```

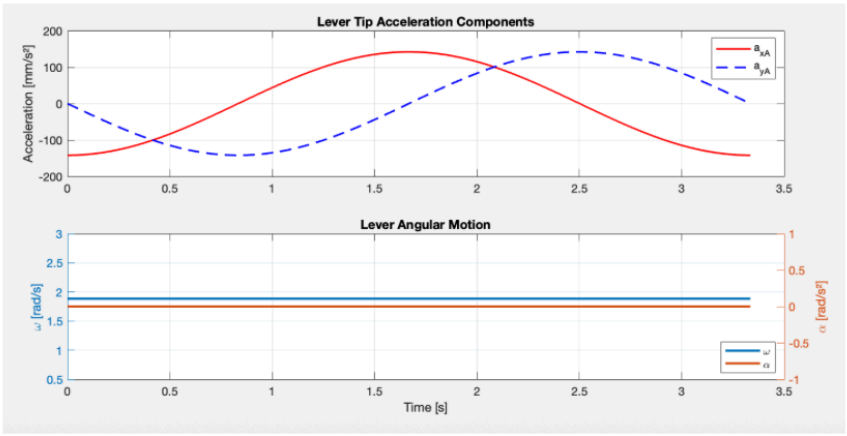
8. MATLAB Analysis

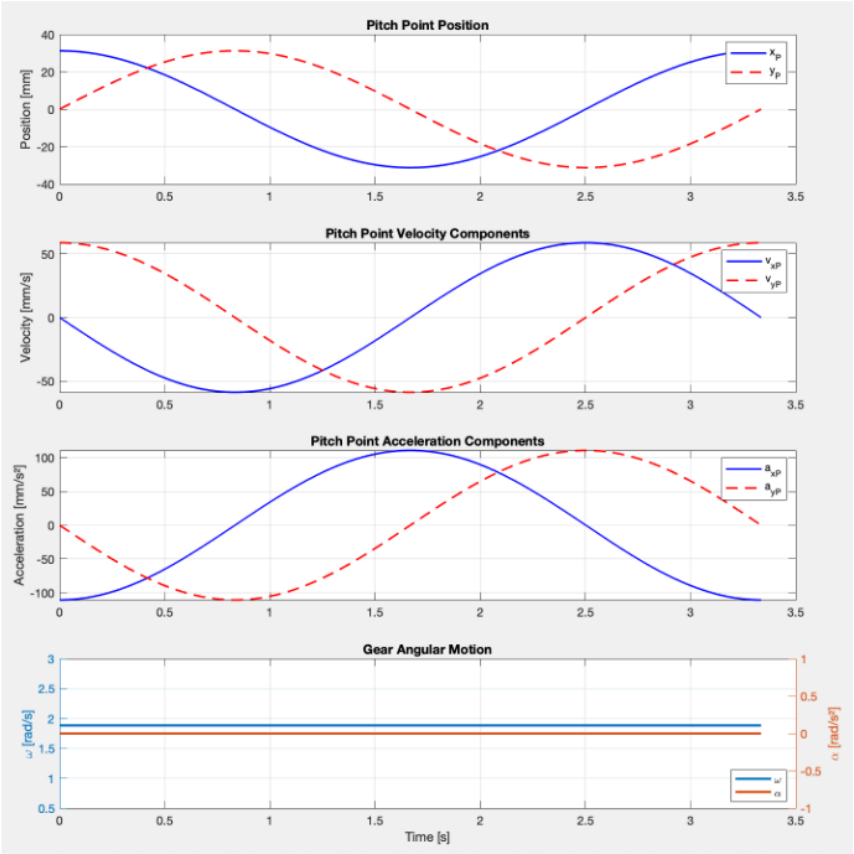
We have established a matlab code that represents our parts starting from giving an initial angular velocity which is our degree of freedom and using the formulas we have in the dynamic analysis part we came up with these graphs and videos. After doing the motion analysis we exported all 3000 data points to an excel sheet and we imported it to the matlab so we can

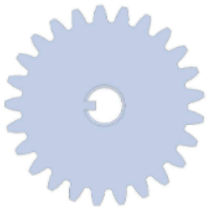
easily compare data from our calculations and Siemens NX motion and later the experimental validation part.



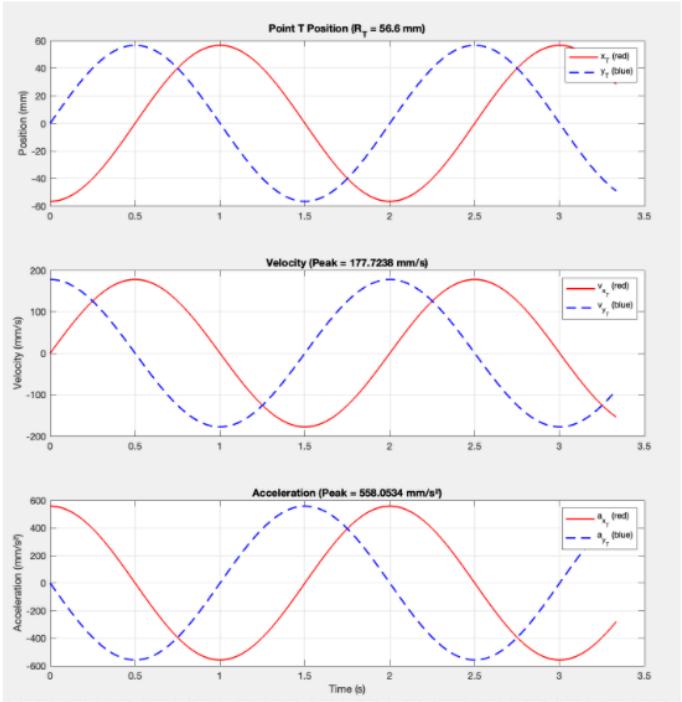


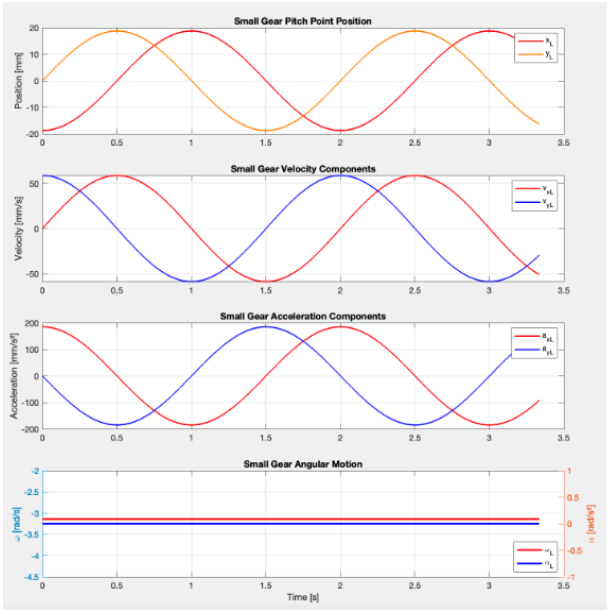


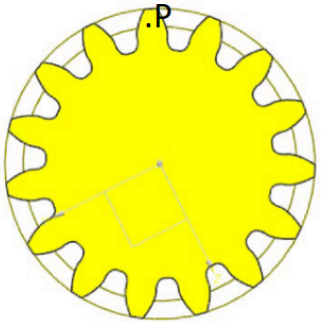

Pitch Point Position
Position [mm]
Pitch Point Velocity Components
Velocity [mm/s]
Pitch Point Acceleration Components
Acceleration [mm/s²]
Gear Angular Motion
ω [rad/s]
α [rad/s²]
Time [s]

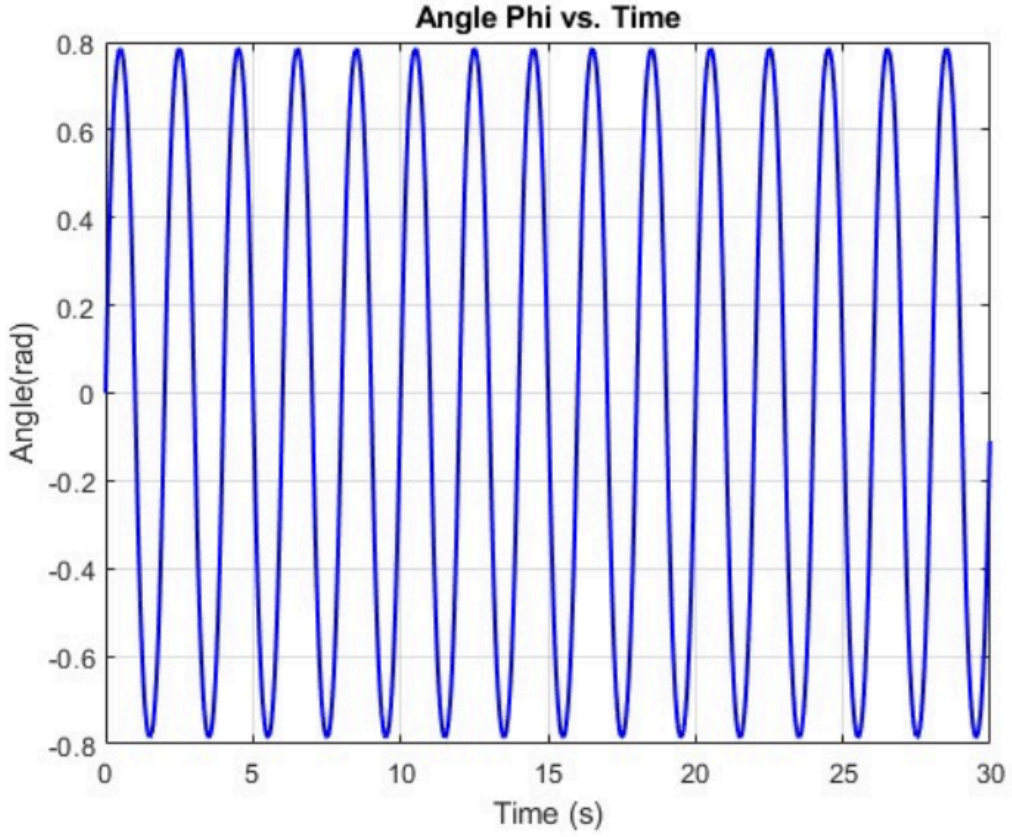


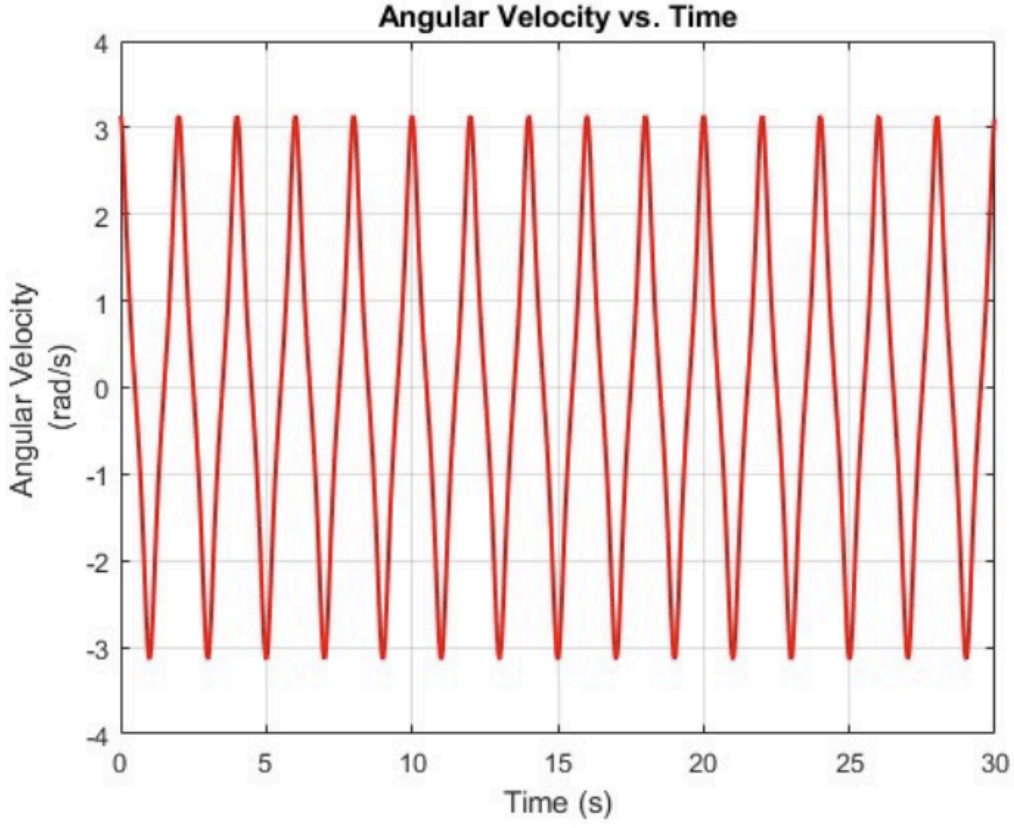


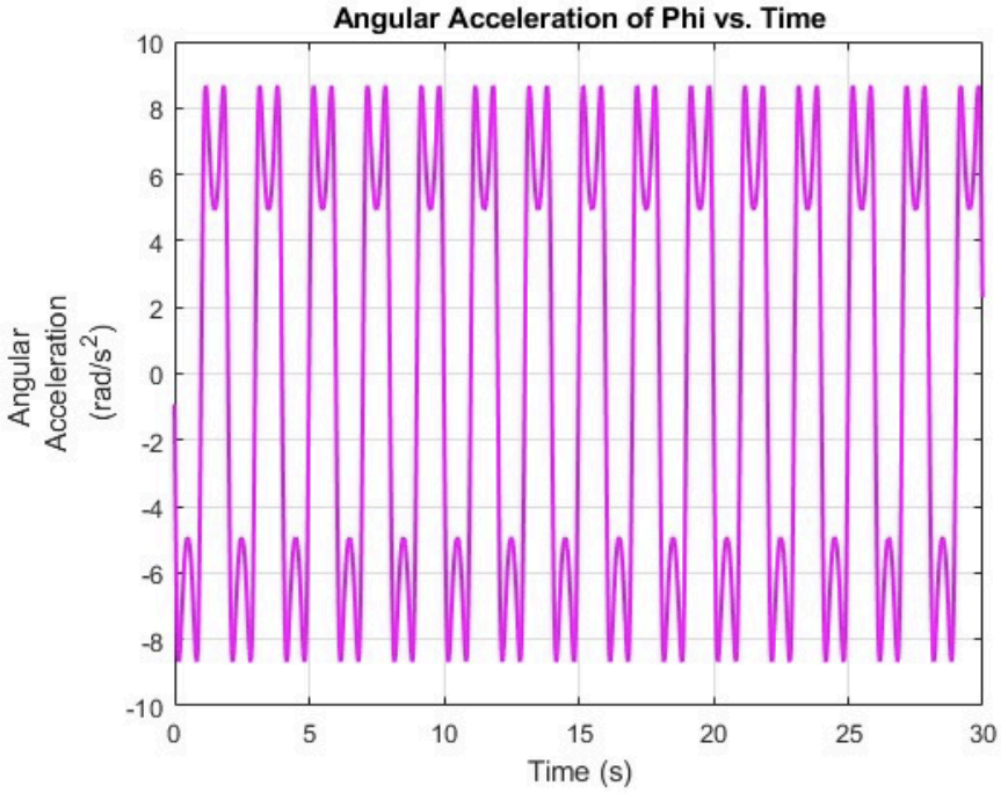

Point T Position (R_T = 56.6 mm)
Position (mm)
Velocity (Peak = 177.7238 mm/s)
Velocity (mm/s)
Acceleration (Peak = 558.0534 mm/s²)
Acceleration (mm/s²)
Time (s)

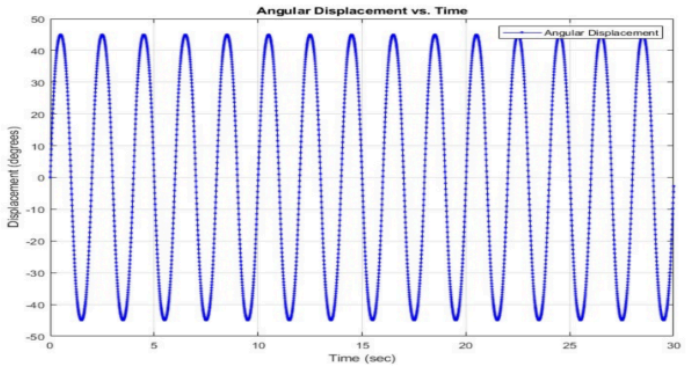

Small Gear Pitch Point Position
Position [mm]
Small Gear Velocity Components
Velocity [mm/s]
Small Gear Acceleration Components
Acceleration [mm/s²]
Small Gear Angular Motion
Time [s]

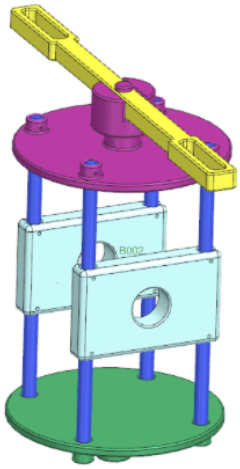

.P


Angle Phi vs. Time
Angle(rad)
Time (s)
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
5
10
15
20
25
30


Angular Velocity vs. Time
Angular Velocity
(rad/s)
Time (s)
4
3
2
1
0
-1
-2
-3
-4
0
5
10
15
20
25
30


Angular Acceleration of Phi vs. Time
Angular Acceleration (rad/s^2)
Time (s)
10
8
6
4
2
0
-2
-4
-6
-8
-10
0
5
10
15
20
25
30


Angular Displacement vs. Time
Angular Displacement
Displacement (degrees)
Time (sec)



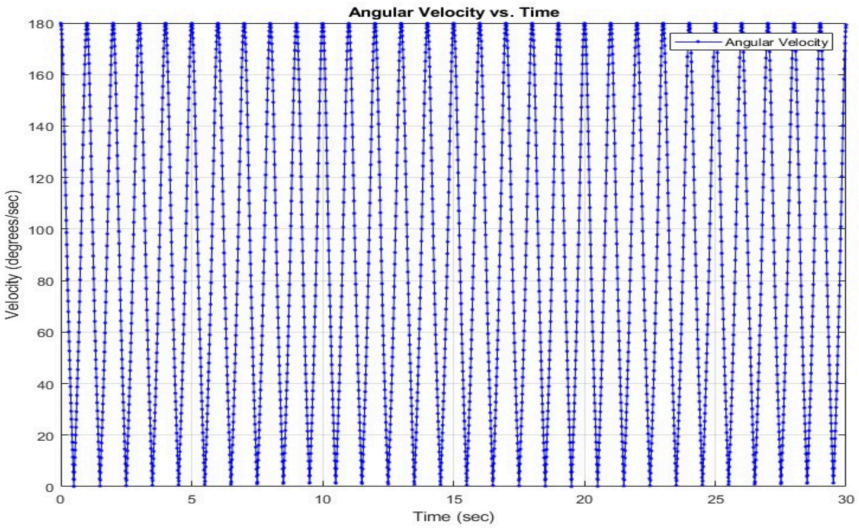
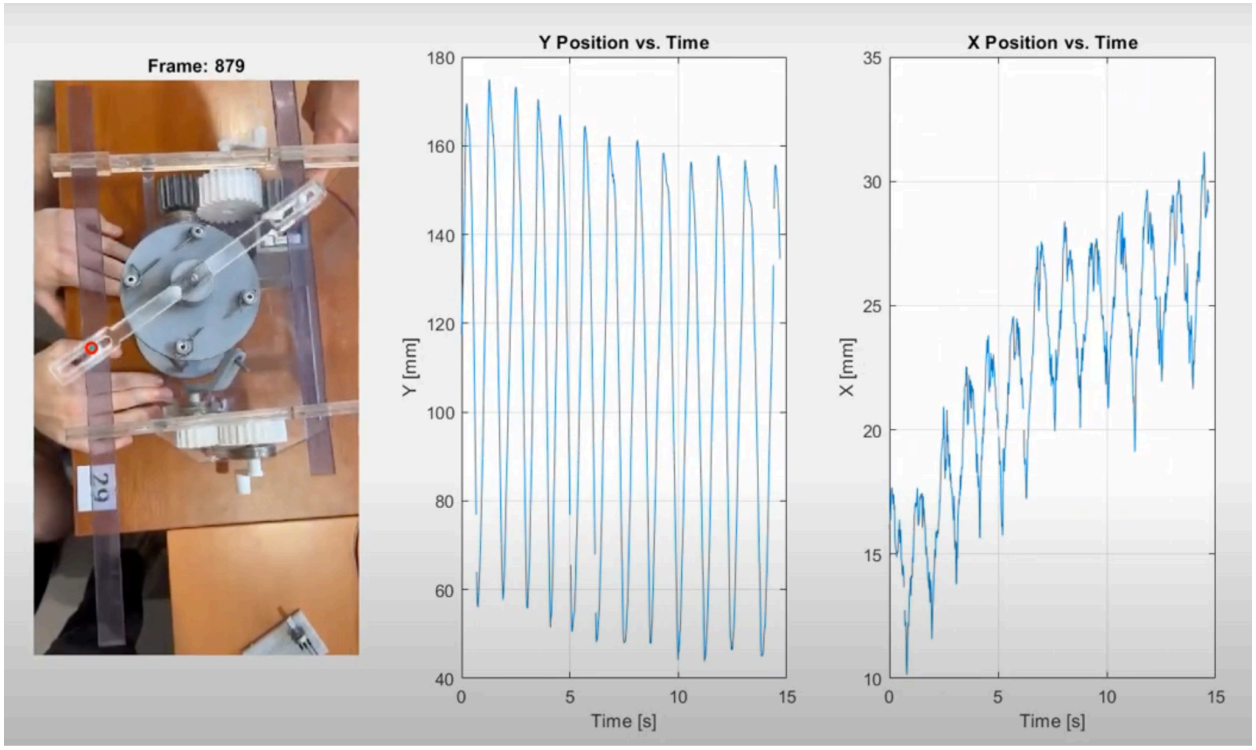


Fig. Angular Displacement and Velocity vs. Time of the Harmonic Middle Part

9. Experimental Validation



For the experimental validation we have run our mechanism and recorded it, at the beginning we had some problems with the tracking because matlab was not able to track the color that we have used then we switched to the green color. After the change the provided

matlab code was able to track the point and generate the graph. Moreover, we had to make some changes in the code in order to eliminate the deviations caused by the vibration of the system. In order to solve this problem we have separated the x and y components of the point and at the end we had one graph for the displacement of the point and the vibration of the point. For the values we have found from the Siemens NX and from the Matlab you can check out the graphs on the part 8 on the Harmonic Motion of the middle part graph.

There are few mismatches between the experimental values and the Matlab values and we can coin the reasons as:

1) There is friction in the experimental validation unlike the matlab.
2) The angle of the video recording might not be the best angle for tracking a point.
3) The flexibility of the material is excluded in the matlab codes but in reality it may cause mismatches.
4) Numerical solutions in the Siemens NX leads to errors.
5) Matlab code is totally about kinematics so the inertia and the mass are neglected unlike real life.

10. Conclusion

The main objective of this project was to design and manufacture a machine that converts rotational motion into linear motion and to perform its dynamic analysis. The mechanism was successfully assembled and operated, with the motor integrated onto a platform and connected to the large gear. The system functioned as expected, effectively producing the desired linear output. Both experimental testing and simulation results confirmed that the dynamic behavior of the mechanism met our performance expectations.

During the project, the team faced challenges such as communication issues and mismatches between the manufactured parts and the original design specifications. These problems were resolved through collaborative meetings and iterative adjustments. For future improvements, the mechanism could be redesigned to achieve a more compact structure, tighter manufacturing tolerances, and optimized material selection to enhance the strength-to-weight ratio. Additionally, advancements in the control system and increased automation could further improve the system's performance and reliability.